Audience-Driven Data Visualization: A Persona-Based Approach

NBA Player Efficiency Analysis (2014-2024)

Mridul Pathak

Virginia Commonwealth University
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How the same analytical findings require different visual approaches for different audiences

Project Context

Role: Independent Research Project (MKTG-678: Marketing Analytics, Spring 2026)
Timeline: January - May 2026 (concurrent with coursework)
Tools: Excel (statistical analysis), NotebookLM (narrative exploration), manual design
Scope: Analyze 2,156 NBA player-seasons (2014-2024) and explore how audience psychographics influence data visualization design choices

Core Question: If the same analytical insight can be framed multiple ways, how do design choices (color, jargon, graph complexity, narrative tone) need to adapt for different audience types?

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1 What This Project Is About

The project brief was straightforward: analyze NBA player data and create one infographic that demonstrates analytical thinking. I did that and completed it. But somewhere between the initial feedback and the final deliverable, I got curious about a different question that wasn't part of the original scope.

The early feedback was helpful—focus on distributions first, then look at bivariate relationships and trends over time. Pretty standard analytical workflow. But one line stood out: *"I wish I could help more, but I'm not very familiar with sports!"*

That got me thinking. If my evaluator—someone who clearly understands data and analytics—felt like he couldn't engage fully with the content because he doesn't follow basketball, then how many other people would look at the same infographic and just...tune out? Not because the analysis is bad, but because the way it's presented assumes a certain baseline knowledge or interest they don't have.

So I started asking: what if I took the exact same dataset and findings, but redesigned the presentation for completely different types of people? Someone who lives on r/NBA versus someone who's never watched a game. A business executive who understands ROI frameworks versus a college student just getting into data visualization. Would the design choices actually need to be different, or could I just "simplify" one version and call it a day?

That question turned into a parallel exploration that ran alongside the main project throughout the semester. The exploratory work helped me figure out what my primary deliverable should prioritize, but it also became interesting on its own.

2 The Core Question

If I have one analytical insight—let's say "player efficiency matters more than raw scoring"—how would I communicate that to four completely different people?

- Someone who follows advanced NBA stats religiously
- A business professional who watches sports casually but doesn't know basketball-specific metrics
- A college student curious about data viz but with zero basketball background
- Someone scrolling social media who doesn't care about sports at all

The standard advice is "make it simpler for beginners" or "add more details for experts." But that treats audience adaptation like a volume knob—just turn the complexity up or down. I suspected it was more than that. Different audiences might need different color schemes, different narrative hooks, different graph types, even different emotional tones.

I wanted to test whether that was actually true.

3 How I Approached This

3.1 Building Audience Personas

I started by mapping out who these different people actually are—not just demographics like age or education, but how they think, what motivates them, how they consume content.

I pulled from a few sources: Pew Research studies on sports fandom, Nielsen reports on digital behavior, and some academic papers on graph literacy (turns out about 40% of adults struggle with basic charts, which was eye-opening).

Here's what I ended up with:

Persona 1: Analytics Enthusiasts

- The people who live on Basketball Reference, follow FiveThirtyEight's RAPTOR ratings, argue about PER vs. Win Shares on Reddit
- They want methodological transparency. They'll actually read your footnotes. They care about outliers and contrarian takes.
- Jargon isn't a barrier for them—it's a signal that the content is serious.

Persona 2: Sports-Aware Generalists

- Business professionals who watch March Madness or follow the NFL, but don't know what "usage rate" means in basketball
- They understand ROI, efficiency metrics, quadrant frameworks—concepts that transfer across sports and business contexts
- They need the analysis framed in language that doesn't require NBA fluency

Persona 3: Curious Newcomers

- People exploring data visualization for the first time, maybe for a class or personal interest
- They need the concept explained before the data. Jargon feels like gatekeeping to them.
- Visual metaphors work better than scatter plots. They want a story, not a statistical test.

Persona 4: Unaware/Uninterested

- People who encounter content passively while scrolling—zero sports interest, zero analytics background
- They need a provocative hook on a universal theme (fairness, recognition, value) not a basketball story
- One big number, minimal text, scroll-stopping design
- I didn't build this one. More on that later.

3.2 The Data Work (Standard Process)

This part was the same across all versions. I used the NBA player-season dataset from Basketball Reference:

- 2,156 player-seasons across 648 unique players
- 10 completed seasons (2014-15 through 2023-24)
- Filtered for players with at least 20 minutes per game (the regulars, not bench warmers)

The analytical findings were consistent no matter who the audience was:

1. Centers have the highest average PER (19.2) despite being the least common position
2. Playoff players average 10.6% higher PER than non-playoff players
3. PPG and PER correlate strongly (r = 0.759), but there are outliers like Jokic who break the pattern
4. Scoring increased 16.6% over the decade

Excel handled most of this—COUNTIF, AVERAGEIF, CORREL, PivotTables. The mechanics weren't complicated.

3.3 Using AI to Explore Narrative Angles

Here's where things got more interesting. After I had the numbers, I needed to figure out how to *frame* them for each persona.

I used NotebookLM for this. I'd upload the Excel outputs along with the persona profile and prompt it with something like: *"Given this efficiency dataset and an audience of business professionals who don't follow basketball, what framing would make them care?"*

What came back wasn't a final product—it was perspective options I wouldn't have thought of on my own:

- For generalists, it suggested an ROI lens: salary vs. contribution, efficiency gaps, economic fairness
- For newcomers, it proposed the "invisible star" metaphor—players who don't score much but make their teammates better
- For enthusiasts, it pointed to Jokic as the key outlier: low usage, disproportionately high impact

I'd then iterate on those ideas, cross-check against my Excel calcs to make sure the narrative held up, and design the actual visuals separately. The AI wasn't making the infographics—it was helping me think through what story each audience would find compelling.

3.4 Documenting Every Design Choice

For each variant, I documented what I chose and why, grounded in actual research rather than just gut feel.

Example: Why did I use orange for the "scoring specialist" in the newcomer version and yellow-green for the "efficiency star"? Because research on color psychology shows warm colors (orange, red) signal dominance and aggression, while yellow-green evokes cooperation and growth. That's not a random aesthetic choice—it's a semantic one. The colors are communicating before the text does.

I ended up with 30+ documented decisions across the three variants—everything from jargon density to graph types to where I placed visual emphasis.

3.5 Project Timeline: Development Process

This exploration happened alongside the primary project deliverable, each feeding into the other:

February 2026: After topic selection and initial dataset assembly, early feedback established the analytical structure: distributions first, then bivariate relationships, then temporal trends. This became the backbone of every version.

March 2026: While refining the core analysis, I built the persona profiles—pulling from research on graph literacy, sports fandom patterns, and business communication norms. The goal was understanding not just *what* to show, but *how* different audiences would process it.

April-May 2026: Used NotebookLM to test narrative angles for each persona. Same data, completely different framing. This revealed that audience adaptation isn't about simplifying or complexifying one design—it's about fundamentally different choices at every level.

Final synthesis: The primary deliverable incorporated insights from all three variants, optimized for sports-aware generalists. The persona exploration directly shaped the final design.

4 The Three Variants I Built

4.1 Variant 1: "The Invisible Star" (For Newcomers)

Goal: Explain the concept of efficiency to someone who's never thought about basketball stats before.

Narrative: "Winning games isn't just about scoring points—it's about making your teammates better. Some players are 'invisible stars' who don't show up on highlight reels but quietly elevate everyone around them."

4.1.1 Design Choices

Color Palette: High-Contrast Orange vs. Yellow-Green

I went with orange for the scorer (30 PTS) and yellow-green for the efficiency player (15 PTS). Research shows warm colors signal dominance and individual performance, while cooler tones evoke collaboration. For someone unfamiliar with basketball, those color associations do some of the work before they even read the text.

I avoided muted tones entirely. Neutral colors would signal "this is complex professional analysis," which would scare off newcomers.

Visual Metaphor: Stars and Silhouettes

Stars are universally understood as "this person is excellent." No basketball knowledge required. The silhouettes keep it abstract—this isn't about specific players, it's about archetypes. That lowers cognitive load (concept from multimedia learning research—people process simple shapes faster than detailed imagery).

I specifically avoided scatter plots or bar charts. Studies show about 40% of adults struggle with basic graph interpretation. If you're trying to welcome someone into data visualization, don't lead with the format that might intimidate them.

The "+20%" Hook

One big number: efficiency players increase their teammates' shooting by 20%. That's the whole message. Heath & Heath's "Made to Stick" framework talks about this—simple, unexpected, concrete messages are memorable. A table with five metrics isn't.

Zero Jargon

I used "efficiency," "multiplier," "invisible impact"—all plain English. For newcomers, technical jargon reads as gatekeeping. They interpret it as "this content isn't for me." I wanted the opposite signal.

Figure 1: Variant 1: "The Invisible Star" — Designed for people with zero basketball knowledge. High-contrast colors (orange vs. yellow-green), metaphorical visuals (stars, silhouettes), one memorable stat (+20%), no jargon. The goal was instant comprehension.

4.2 Variant 2: "ROI on the Court" (For Generalists)

Goal: Make the analysis accessible to business-minded people who follow sports but not NBA-specific metrics.

Narrative: "Which players deliver the best value for their salary? Let's look at this like an investment portfolio."

4.2.1 Design Choices

Color Palette: Tan and Earth Tones

The tan background, brown and navy dots—this was deliberate. I wanted it to feel like a financial document, not a sports entertainment piece. Neutral earth tones signal professionalism and analytical rigor. It's the visual equivalent of the Wall Street Journal aesthetic.

I avoided bright team colors (Lakers purple, Celtics green) because that would make it feel tribal. Generalists aren't invested in specific teams—they want objective analysis.

Scatter Plot with Quadrant Framework

X-axis is salary, Y-axis is Win Shares (contribution to team success). Reference lines split it into quadrants: high cost/low output, high cost/high output, low cost/low output, low cost/high output.

This structure is familiar from the BCG Growth-Share Matrix, which most business school grads have seen. Quadrant thinking is intuitive for this audience—they immediately understand "top right is good."

I skipped regression lines and confidence intervals. Too technical for people who aren't stats-focused.

Faint Basketball Court Outline

There's a subtle court shape in the background. It's not prominent enough to demand basketball expertise, but it's enough to activate what's called a "perceptual schema"—your brain sees the court and goes "okay, this is sports-related, I can engage with this."

Business Terminology, Not NBA Jargon

"ROI," "salary efficiency gap," "gold standard anchors," "league average"—these are all terms that transfer across sports and business contexts. I avoided PER, RAPTOR, usage rate. Those are insider metrics.

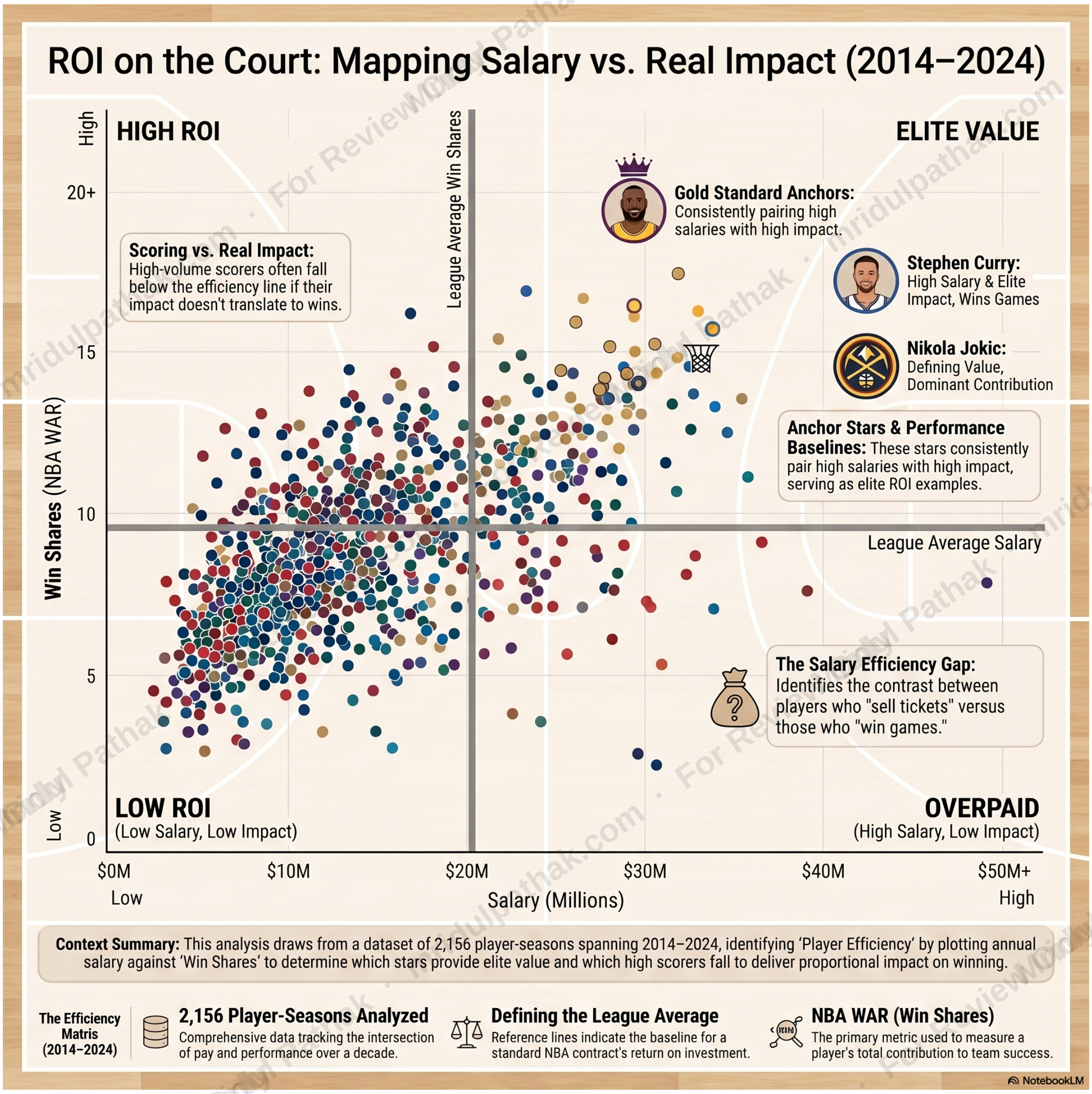

Figure 2: Variant 2: "ROI on the Court" — Designed for business professionals. Muted earth tones, scatter plot with quadrant framework (familiar from BCG matrix), faint court outline as context, business-framed language. Bridges sports analytics and business strategy.

4.3 Variant 3: "Efficiency vs. Volume" (For Enthusiasts)

Goal: Give analytics nerds something they can't get from a generic sports site—methodological rigor and contrarian insights.

Narrative: "Does the efficiency-volume trade-off actually hold? Let's look at the outliers who break the model."

4.3.1 Design Choices

Color Gradient Encoding RAPTOR Score

The dots aren't just random colors—they encode a third variable (RAPTOR Impact Score). Cool tones (navy, purple) represent low impact, warm tones (burgundy, gold) represent high impact. This follows ColorBrewer guidelines for sequential data.

Enthusiasts expect data density. The gradient adds a dimension without cluttering the chart.

Scatter Plot + Regression Line + Confidence Interval

This tests a well-known hypothesis in basketball analytics: as usage increases, efficiency typically declines (you can't maintain peak performance when you're taking 30 shots a game).

The dashed regression line shows the expected relationship. The shaded confidence band shows statistical uncertainty. Outliers above the line—like Jokic—are the interesting cases.

I kept the confidence interval because this audience audits methodology. If you don't show uncertainty, they'll question your rigor.

Player-Specific Callouts

Jokic (2020-21): Low usage, disproportionately high RAPTOR. This is the famous efficiency-volume paradox case.

Curry (2015-16): ¿30% usage, still 5-8% above adjusted efficiency mean. The 73-9 Warriors season where he broke the model.

Enthusiasts care about *who* breaks the pattern, not just that the pattern exists.

Advanced Metrics and Methodology Notes

"ERA-Adjusted True Shooting %," "RAPTOR Impact Score," "Regression Framework," "Sample Size Requirements"—this is the language this audience expects. Jargon validates that the content is for them. If I wrote in plain English, they'd dismiss it as pop analytics.

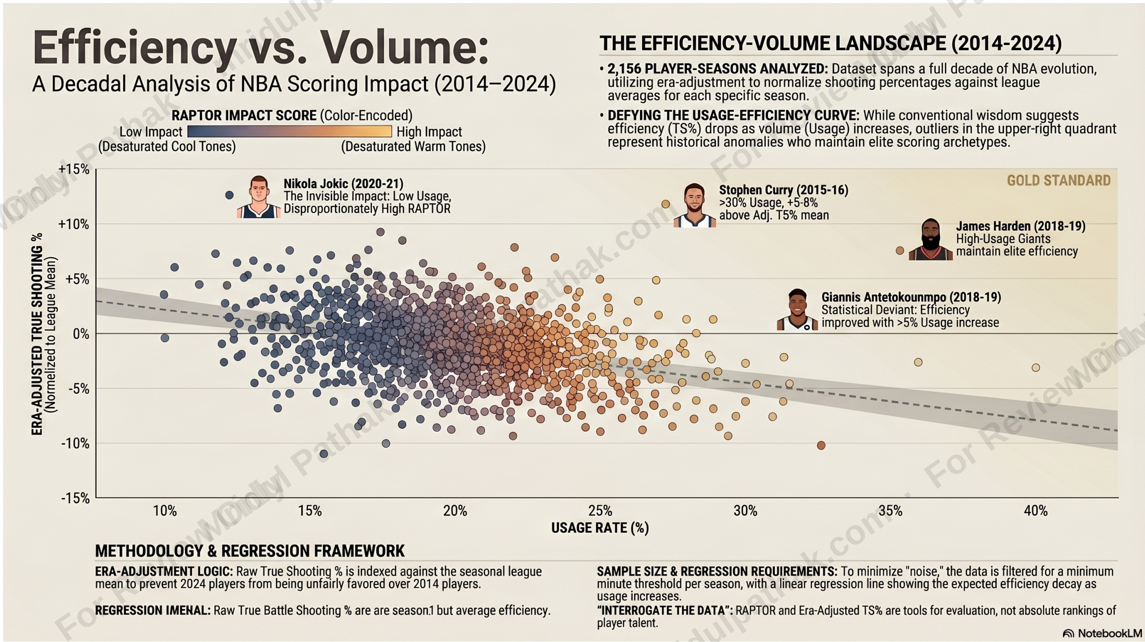

Figure 3: Variant 3: "Efficiency vs. Volume" — Designed for analytics enthusiasts. Sequential color gradient (encoding RAPTOR), scatter plot with regression + confidence interval, specific player callouts (Jokic, Curry, Harden, Giannis), advanced metrics, methodology documentation. Prioritizes statistical rigor and contrarian insights.

5 Why I Didn't Build the Fourth Variant

The fourth persona—people with zero sports interest who encounter content while scrolling—would require a completely different validation approach.

Based on research, it would need:

- Near-monochrome design with one neon accent (for scroll-stopping contrast)
- One massive "hero number" and fewer than 50 words total
- A provocative, non-sports headline like "Is This America's Most Underpaid Specialist?"

- Universal human themes (fairness, value, recognition) instead of basketball framing

But here's the problem: creating scroll-stopping content for disinterested audiences requires metrics I don't have access to. Click-through rates on different headline styles. Dwell time on minimal vs. detailed layouts. Sharing behavior.

The other three variants are grounded in actual audience research—surveys on sports fans, graph literacy studies, business communication norms. The fourth would be guesswork. I'd rather skip it than fake it.

6 Primary Deliverable: Generalist-Optimized Design

Based on the persona exploration, I optimized the primary deliverable for sports-aware generalists—people comfortable with analytical thinking but unfamiliar with domain-specific jargon.

Design Specifications:

- **Navy background with structured bar charts:** Professional aesthetic, not entertainment-focused
- **Plain-English metric definitions upfront:** "PPG Points Per Game — avg points scored per game per season." No assumed knowledge.
- **Structured sections following analytical workflow:** Univariate distributions → Temporal trends → Bivariate relationships → Performance rankings
- **Moderate jargon with context:** Uses terms like "Pearson r = 0.759" but explains it as "a strong positive relationship"
- **Data transparency throughout:** Sample size (n = 2,156), filter criteria (20 MPG), source citation all visible

How It Synthesized All Three Variants:

From the newcomer variant: Plain-English definitions, narrative section headers, conversational framing that doesn't assume basketball expertise

From the generalist variant: Professional color scheme, business-appropriate chart types (bar charts, scatter plots, tables), moderate jargon with explanations

From the enthusiast variant: Data transparency (methodology notes, sample size), specific player callouts (Jokic, Embiid, Curry), correlation coefficients documented

Technical Implementation:

The visual tools demonstrate standard analytical techniques:

- Bar charts: COUNTIF, AVERAGEIF for categorical breakdowns
- Line charts: Time-series analysis with seasonal averaging
- Scatter plot: CORREL for bivariate relationships
- Data table: Conditional formatting and ranking

The final design balanced accessibility (definitions, narrative hooks) with analytical rigor (transparency, statistical validation). That hybrid approach came directly from testing how different audiences engage with the same data.

7 What I Learned from This

7.1 Color Isn't Just Decorative

Different audiences read color palettes as signals before they process the actual content:

- High-contrast primaries (orange, yellow, blue) = "this will be easy to follow"
- Muted earth tones (tan, brown, olive) = "this is professional analysis"
- Sequential gradients (cool to warm) = "this has statistical depth"

If there's a mismatch—like playful colors paired with dense regression analysis—it creates cognitive dissonance. The design is sending mixed signals.

7.2 Jargon Works Like an Audience Filter

Technical terminology doesn't just convey information. It signals who the content is for.

For experts, jargon validates that you respect their expertise. For novices, it reads as "this isn't for you." You can't solve that by adding a glossary to a single version. You need separate designs.

7.3 The Same Data Tells Different Stories

My NBA dataset supports all of these narratives simultaneously:

1. "Invisible stars make teammates better" (newcomer framing)
2. "Salary efficiency reveals market gaps" (generalist framing)
3. "Jokic defies the efficiency-volume trade-off" (enthusiast framing)

Effective communication isn't about finding *the* story in the data. It's about choosing the story that resonates with *this specific audience*.

7.4 AI as an Iteration Tool

The value of NotebookLM wasn't producing final outputs. It was compressing weeks of brainstorming into hours.

It suggested the ROI framing for generalists when I was stuck thinking in terms of PER. It proposed the "invisible impact" metaphor for newcomers. It identified Jokic as the key outlier for enthusiasts.

Best use case: perspective generation, not content generation. I'd iterate with it, then validate and build the visuals myself.

8 Where This Framework Applies Beyond Basketball

This isn't just about sports analytics. The persona-driven approach works anywhere you need to communicate the same information to different audiences.

8.1 Product Strategy Internship Application (Summer 2026)

This summer, I'm working as a Product Strategy intern at a marketing infrastructure agency in Richmond. Part of the role involves creating dashboards and reports for two distinct client types using the same platform data:

- **Content creators:** These users care about engagement metrics—views, watch time, audience demographics. For them, dashboards need simplified visualizations with actionable takeaways ("Your best posting time is 7 PM on Tuesdays"). They want recommendations, not raw numbers.
- **Local businesses:** These users care about conversion metrics—leads generated, cost-per-acquisition, ROI on marketing spend. They need business-framed language, competitive benchmarking, and budget efficiency analysis.

Same backend data infrastructure. But the frontend presentation—what metrics surface first, what language describes them, what level of visual complexity matches the user's sophistication—needs to adapt based on who's logging in.

The persona framework from this project is directly informing how I'm approaching that work. It's not theoretical anymore—it's how I'm deciding which dashboard modules to build for which user type.

8.2 Research Communication

Same study, three outputs:

- Academic paper: Full methodology, regression tables, p-values
- Industry whitepaper: Executive summary, case examples, actionable takeaways
- Public blog: Plain-English summary, visual metaphors, real-world analogies

8.3 MAT: Idea Validation Platform (In Development)

I'm building a research aggregation tool that helps people validate business ideas. The core challenge is that the same market intelligence needs to be presented differently depending on where someone is in their ideation process:

- **Early exploration phase:** Users at this stage are exploring whether an idea has potential at all. They need high-level market sizing, trend snapshots, and comparable case studies. The goal is to build confidence and reduce friction—too much detail overwhelms and paralyzes decision-making.
- **Validation phase:** Users here are stress-testing the idea against reality. They need competitive landscape breakdowns, customer segmentation data, and financial projection frameworks. This is where detailed analysis actually helps—they're ready for complexity because they're trying to make an informed go/no-go decision.
- **Pitch preparation:** Users at this stage are packaging the idea for investors or stakeholders. They need curated statistics formatted for slide decks, visual charts optimized for presentations, and risk/opportunity matrices that tell a persuasive story.

Same research API serving all three stages. But the UI adapts complexity, detail depth, and narrative tone based on detected user intent—essentially applying this project's persona-driven visualization framework to information architecture.

The question I'm working through is: how do you detect intent without making the user fill out a survey? Right now I'm testing behavioral signals (how quickly they're moving, whether they're downloading vs. just browsing, what questions they're asking) to infer which stage they're in and adapt the presentation accordingly.

9 Limitations and What I'd Do Differently

9.1 No User Testing

I built the personas from secondary research—Pew studies, Nielsen reports, academic papers on graph literacy. I didn't test these variants with actual users.

If I were doing this properly, I'd A/B test with persona-matched audiences and measure:

- Comprehension (can they accurately summarize the finding after viewing?)
- Engagement (how long do they spend with it?)
- Trust (do they find it credible?)
- Recall (what do they remember a week later?)

9.2 Single Domain

I only tested this on sports data. Color psychology research, jargon tolerance, visual literacy patterns might work differently for healthcare metrics, financial data, environmental statistics. Generalization is an open question.

9.3 AI Tool Constraints

NotebookLM's design outputs limited the stylistic range. A future version would compare AI-generated variants against what professional designers would create for the same personas. Do they make similar audience-driven choices? Different ones?

10 Final Thoughts

This started as a practical question: what should my primary deliverable prioritize? The early feedback suggested focusing on distributions, then bivariate relationships, then trends. That gave me a structural roadmap.

But the comment "I'm not very familiar with sports" made me realize the structure wasn't the only variable. The same structure could work for multiple audiences if the design choices—color, jargon, visual complexity, narrative framing—adapted to who was looking at it.

The exploration became interesting beyond just "how should I complete this project." It turned into a framework for thinking about how visual communication changes based on audience psychographics.

The primary deliverable incorporated insights from all three variants. But the variants themselves revealed something broader: effective data visualization isn't about finding one perfect design. It's about understanding who you're designing for and what they need the visualization to do for them.

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